

Motivating exact matching via read mapping

CMSC701

(Short) Read mapping/alignment

Read mapping / alignment is one of the most fundamental computational tasks in genomics.

Performing read mapping is often the first step in *many* different analyses.

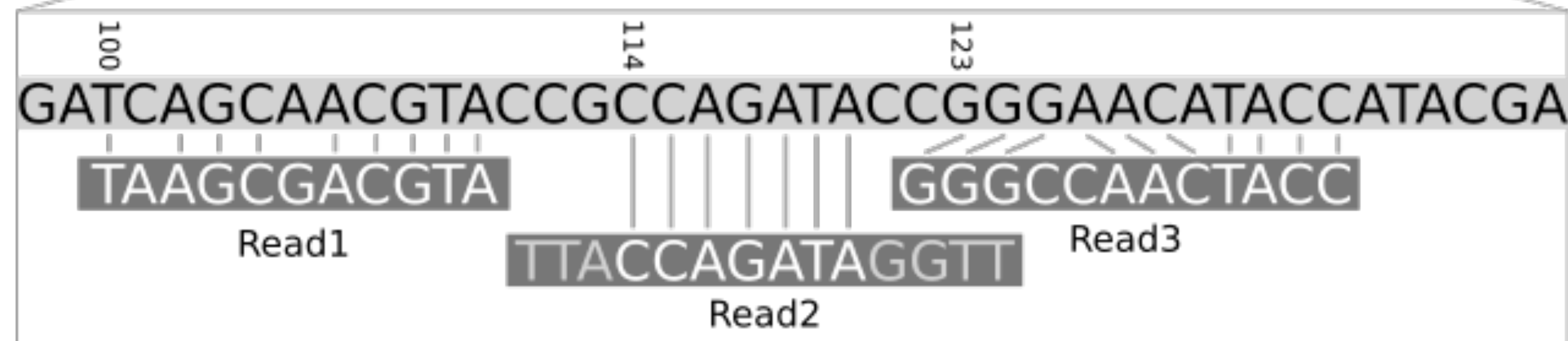
Q: For each **read**, where might I have sequenced it from on the **genome**, and how does the **read** differ from the **reference** at the mapped position?

(Short) Read mapping/alignment

Set of reads

Reference genome

Mapping



Short reads:

10^6 - 10^9 reads

100-300 nt per end

Often “paired-end”

Genome:

10^6 - 10^{10} nt long

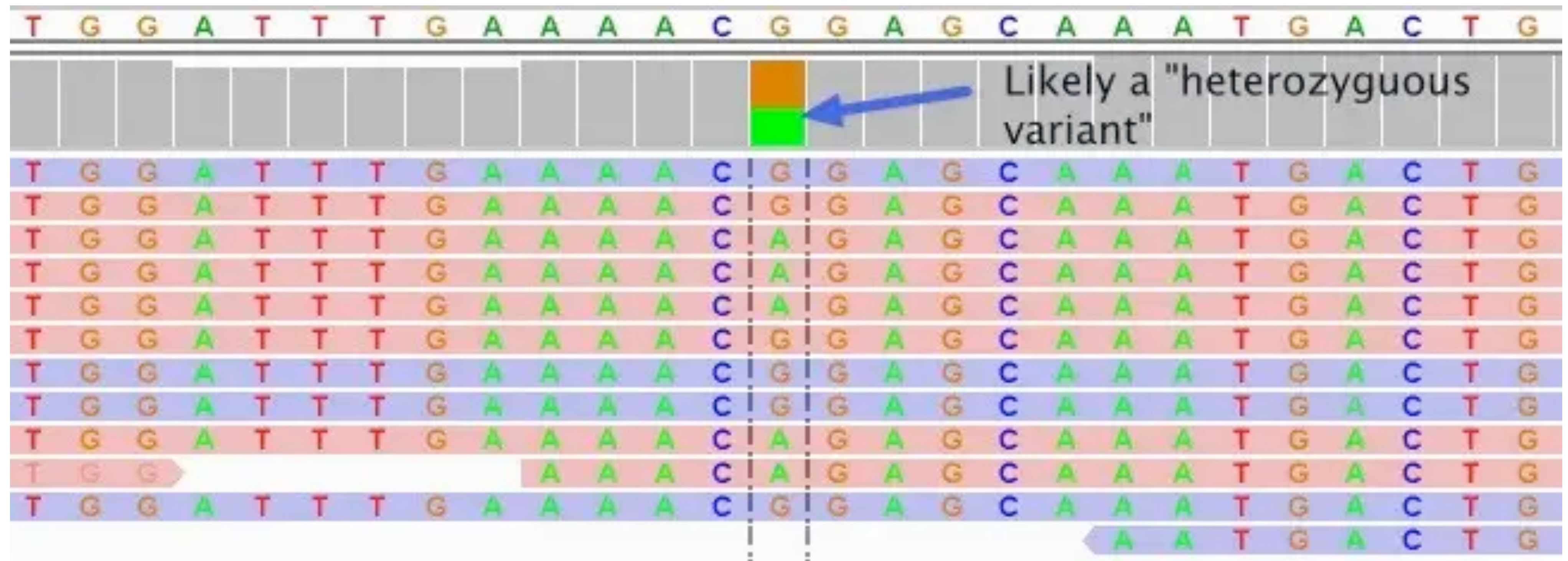
May contain gaps / Ns

Q: For each **read**, where might I have sequenced it from on the **genome**?

Read mapping → variant calling

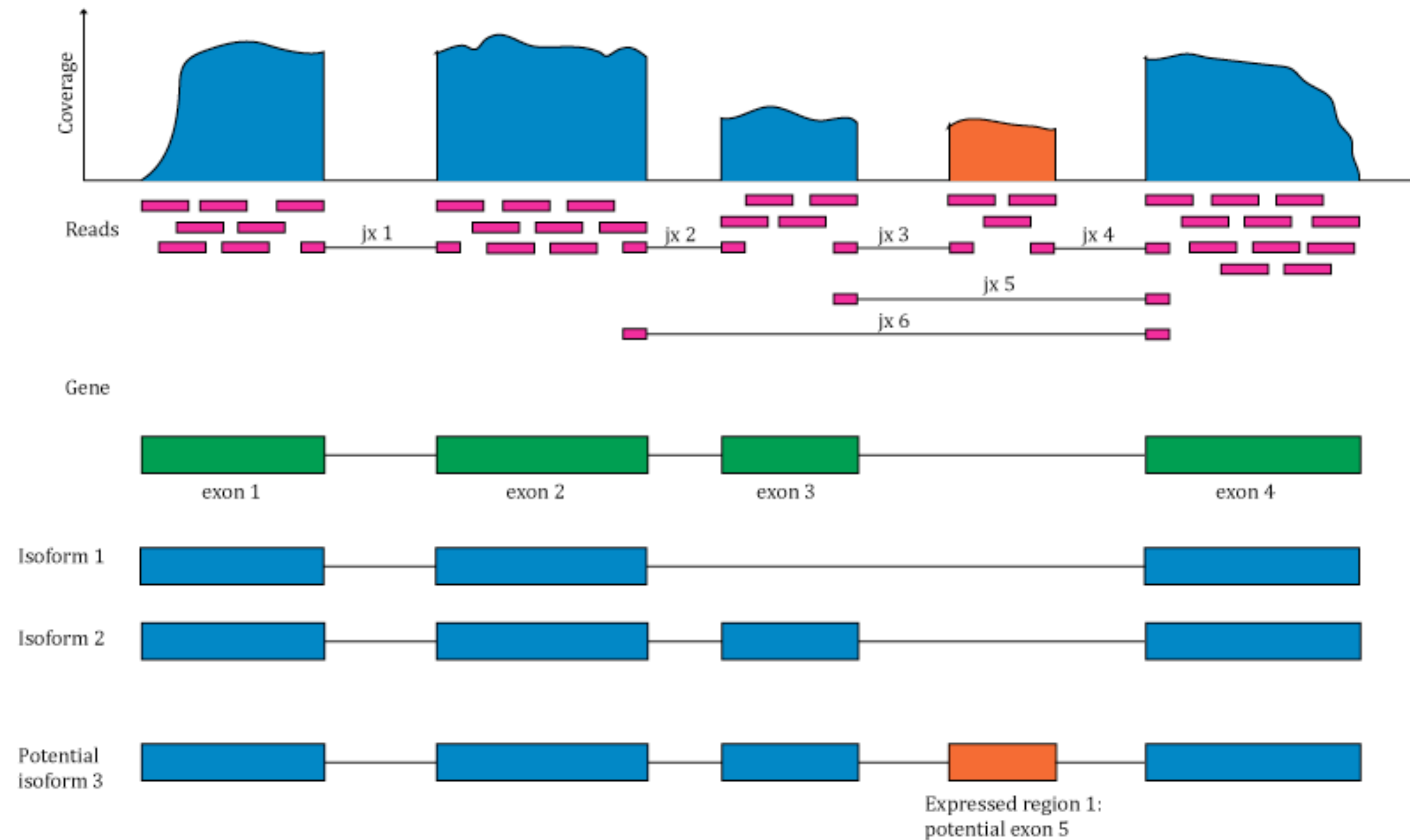
Reference genome

Mapped reads from sample



Given the alignment of many reads to the “reference”, how and where does my sample differ from the reference?

Read mapping → count/census



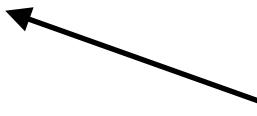
Given a sequencing sample, which genes (isoforms) are generating reads, and how many reads are coming from each (quantitative measure of expression level).

The utility of *exact* matching here

As *loose* motivation, consider the problem of mapping a read r to the genome G .

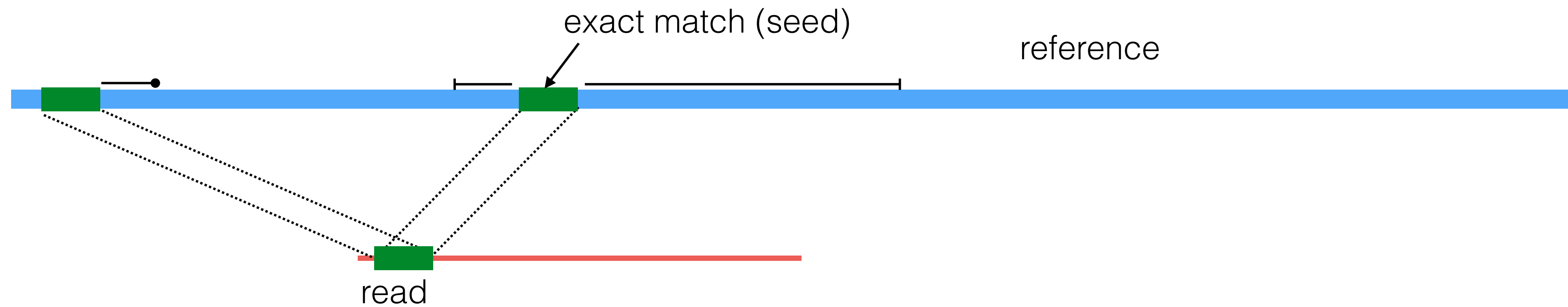
In reality, we would not use exact matching for this; why?

However, exact matching is useful here:

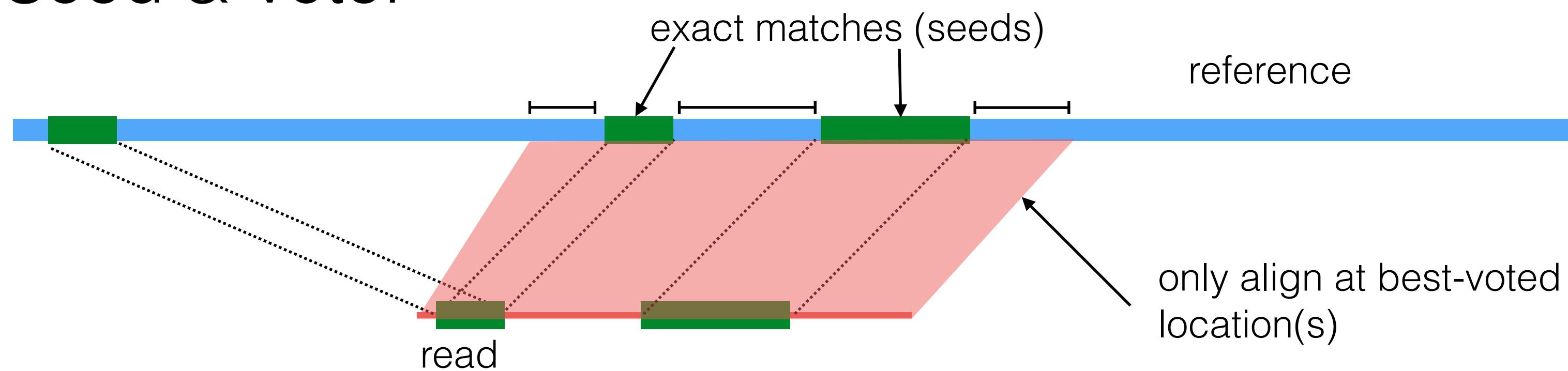
- Find all places where a substring of the query matches the reference exactly (seeds)  Requires efficient exact search
- Filter out regions with insufficient exact matches to warrant further investigation
- Perform a “constrained” alignment that includes these exact matching “seeds”  Here is where we use efficient algorithms for inexact matching (alignment)

Typical Strategies

Seed & Extend:



Seed & Vote:



Representing alignments

@HD VN:1.5 SO:coordinate											Header section
@SQ SN:ref LN:45											
r001	99	ref	7	30	8M2I4M1D3M	=	37	39	TTAGATAAAGGATACTG	*	Alignment section
r002	0	ref	9	30	3S6M1P1I4M	*	0	0	AAAAGATAAGGATA	*	
r003	0	ref	9	30	5S6M	*	0	0	GCCTAAGCTAA	* SA:Z:ref,29,-,6H5M,17,0;	
r004	0	ref	16	30	6M14N5M	*	0	0	ATAGCTTCAGC	*	
r003	2064	ref	29	17	6H5M	*	0	0	TAGGC	* SA:Z:ref,9,+,5S6M,30,1;	
r001	147	ref	37	30	9M	=	7	-39	CAGCGGCAT	* NM:i:1	

What is a CIGAR string?

```
RefPos:      1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19
Reference:    C  C  A  T  A  C  T  G  A  A  C  T  G  A  C  T  A  A  C
Read:  ACTAGAATGGCT
```

Aligning these two:

```
RefPos:      1  2  3  4  5  6  7      8  9 10 11 12 13 14 15 16 17 18
Reference:    C  C  A  T  A  C  T      G  A  A  C  T  G  A  C  T  A  A
Read:          A  C  T  A  G  A  A      T  G  G  C  T
```

With the alignment above, you get:

```
POS: 5
CIGAR: 3M1I3M1D5M
```

What is a CIGAR string?

6. CIGAR: CIGAR string. The CIGAR operations are given in the following table (set ‘*’ if unavailable):

Op	BAM	Description	Consumes query	Consumes reference
M	0	alignment match (can be a sequence match or mismatch)	yes	yes
I	1	insertion to the reference	yes	no
D	2	deletion from the reference	no	yes
N	3	skipped region from the reference	no	yes
S	4	soft clipping (clipped sequences present in SEQ)	yes	no
H	5	hard clipping (clipped sequences NOT present in SEQ)	no	no
P	6	padding (silent deletion from padded reference)	no	no
=	7	sequence match	yes	yes
X	8	sequence mismatch	yes	yes

- “Consumes query” and “consumes reference” indicate whether the CIGAR operation causes the alignment to step along the query sequence and the reference sequence respectively.
- H can only be present as the first and/or last operation.
- S may only have H operations between them and the ends of the CIGAR string.
- For mRNA-to-genome alignment, an N operation represents an intron. For other types of alignments, the interpretation of N is not defined.
- Sum of lengths of the M/I/S/=/X operations shall equal the length of SEQ.